

Instructions: Answer all questions. Write in complete sentences unless told otherwise, and show your work for calculations.

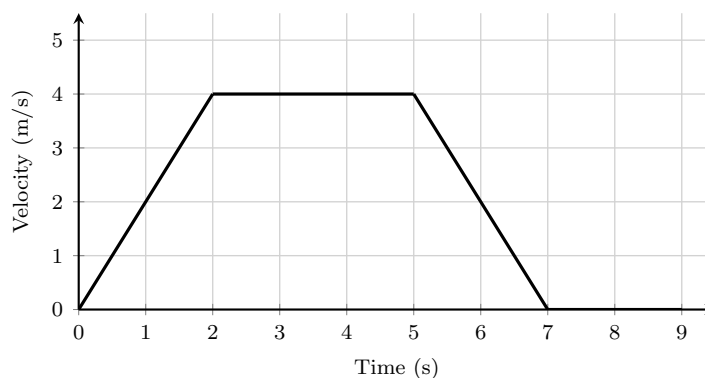
Area Under a Velocity–Time Graph

The **area** between the line and the time axis equals the **displacement**.

$$\text{rectangle : } A = bh \quad \text{triangle : } A = \frac{1}{2}bh \quad \text{trapezoid : } A = \left(\frac{h_1 + h_2}{2}\right)b \text{ or } \left(\frac{b_1 + b_2}{2}\right)h$$

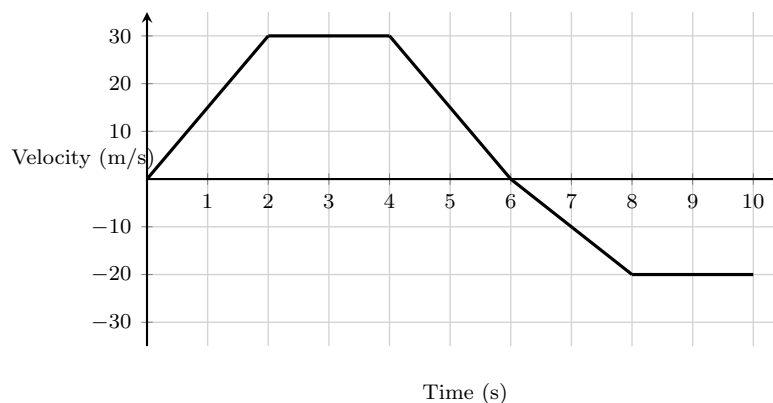
- Split the area into **rectangles**, **triangles**, and **trapezoids**.
- Area **above** the time axis is **positive** displacement. Area **below** the time axis is **negative** displacement.
- Add the positive and negative areas to get the **total displacement**.

1. The velocity–time graph below shows the motion of a cyclist along a straight path. The graph is made of four straight segments.



- (a) During which time interval(s) is the cyclist speeding up?
- (b) During which time interval(s) is the cyclist slowing down?
- (c) During which time interval(s) does the cyclist have a constant velocity?
- (d) During which time interval(s) is the cyclist at rest?
- (e) Calculate the cyclist's displacement from $t = 0$ s to $t = 9$ s by finding the area under the graph.

2. The velocity–time graph below shows the motion of a remote-controlled car along a straight track. The graph is made of five straight segments.



- (a) At what time does the car change direction?
- (b) During which time interval(s) is the car speeding up?
- (c) During which time interval(s) is the car slowing down?
- (d) During which time interval(s) does the car have a constant velocity?
- (e) Calculate the car's displacement from $t = 0$ s to $t = 6$ s by finding the area under the graph.
- (f) Calculate the car's displacement from $t = 6$ s to $t = 10$ s by finding the area under the graph.
- (g) What is the car's total displacement for the whole trip ($t = 0$ s to $t = 10$ s)?